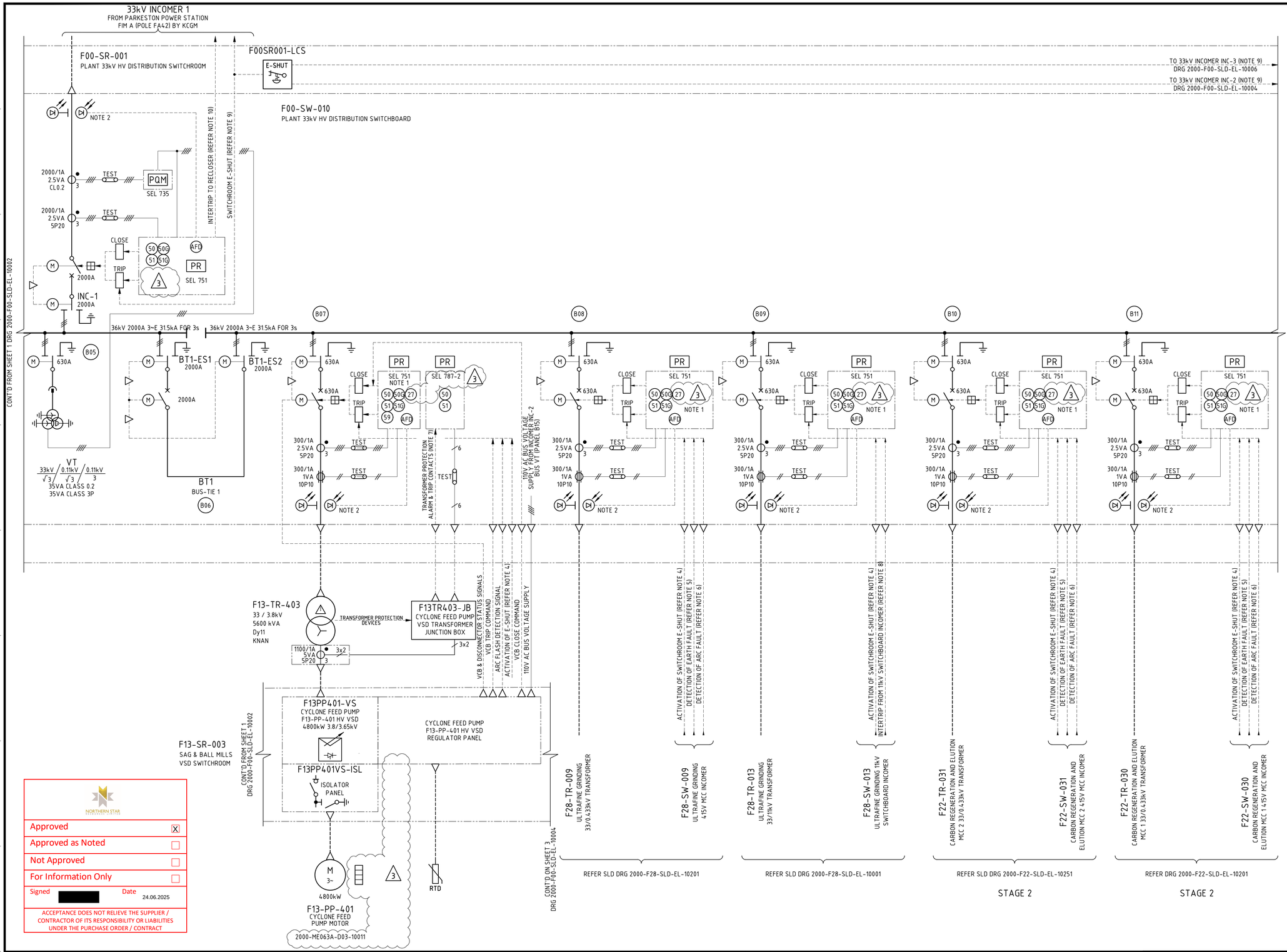


NOTES

1. THE FEEDER PROTECTION RELAY VOLTAGE INPUT SHALL BE WIRED FROM THE CORRESPONDING INCOMER BUS VT.
2. ARC FLASH POINT DETECTORS SHALL BE PLACED IN CABLE ZONES AND CONNECTED DIRECTLY TO THE SEL PROTECTION RELAYS IN EACH PANEL.
3. SWITCHBOARD SHALL BE EXTENSIBLE FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER PER BUS SECTION AS SHOWN.
4. (3) TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE ACTIVATION OF THE SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP THE CIRCUIT BREAKER.
5. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE EARTH FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
6. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE ARC FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
7. TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS SHALL BE HARDWIRED TO THE SEL787 TRANSFORMER FEEDER PROTECTION RELAY.
8. THE INTERTRIP OF THE 33kV FEEDER SHALL BE HARDWIRED FROM THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY EITHER ON A TRANSFORMER TRIP, EARTH FAULT DETECTION OR ARC FAULT DETECTION.
9. (3) THE ACTIVATION OF THE PLANT 33KV DISTRIBUTION SWITCHROOM F00-SR-001 EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP ALL THREE INCOMER CIRCUIT BREAKERS AND THE CORRESPONDING UPSTREAM POLE MOUNTED RECLOSERS.
10. PROTECTION TRIP FROM THE INCOMER CIRCUIT BREAKER SHALL INTERTRIP THE UPSTREAM POLE MOUNTED RECLOSER.

LEGEND

- PR — INCOMER / FEEDER PROTECTION RELAY NOTES 1 & 2
- PQM — POWER QUALITY AND REVENUE METERING
- ARC FLASH SENSOR (POINT TYPE) NOTE 2
- CAPACITIVE VOLTAGE INDICATION
- MECHANICAL OR ELECTRICAL INTERLOCK DEVICE
- (27) UNDER-VOLTAGE
- (50) INSTANTANEOUS OVERCURRENT
- (50G) (50N) INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
- (51) TIMED OVERCURRENT
- (51G) (51N) TIMED EARTH/NEUTRAL OVERCURRENT
- (59) OVER VOLTAGE
- (64) EARTH FAULT
- (67) (67G) DIRECTIONAL OVERCURRENT/EARTH FAULT
- (AFD) ARC FLASH DETECTION

NORTHERN STAR RESOURCES LTD

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 24.06.2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

DRG No.	REFERENCES
2000-F00-DRG-EL-10001	F00-SR-001 33kV PLANT HV DISTRIBUTION SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F00-DRG-EL-10052	F00-SW-010 33kV PLANT HV DISTRIBUTION SWITCHBOARD INTERLOCKING SLD SHEET 2 OF 5
2000-F00-SLD-EL-10004	F00-SW-010 33kV PLANT HV DISTRIBUTION SWITCHBOARD SINGLE LINE DIAGRAM SHEET 3
2000-F00-SLD-EL-10002	F00-SW-010 33kV PLANT HV DISTRIBUTION SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1
2000-F00-SLD-EL-10001	OVERALL SITEWIDE SINGLE LINE DIAGRAM
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION
3	30MAY25	FAT REDLINES INCORPORATED
2	24MAY24	TIER ORDER REVISED.
1	05APR24	TIER ORDER REVISED. REISSUED FOR CONSTRUCTION
0	07MAR24	APPROVED FOR CONSTRUCTION

DRAW	CHECK	DESIGN	TECH APP	PROJ APP
[redacted]				

NORTHERN STAR RESOURCES LIMITED

PRIMERO
primero.com.au | DESIGN | CONSTRUCT | OPERATE

PRIMERO PROJECT NUMBER: 23517

INFORMATION AND REPRESENTATIONS ON THIS DRAWING ARE THE INTELLECTUAL PROPERTY OF PRIMERO UNLESS SPECIFICALLY IDENTIFIED OTHERWISE. ALL RIGHTS INCLUDING USE RESERVED

PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: GENERAL FIMISTON SITE
F00-SW-010 33kV SWITCHBOARD
SINGLE LINE DIAGRAM SHEET 2

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F00-SLD-EL-10003	3

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F00 - GENERAL FIMISTON SITE\2000-F00-SLD-EL-10003.DWG 03JUN25 8:45AM SANDRA CULL